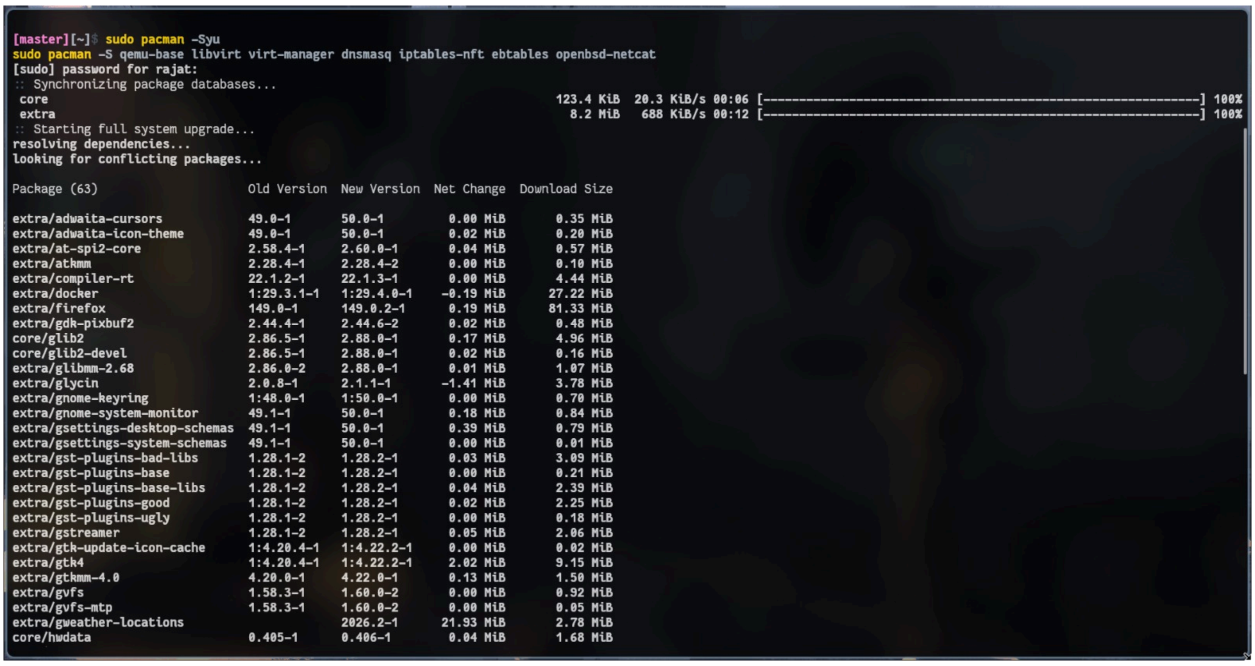
Installation and Configuration of virtualization using KVM.

COMMANDS FOR KVM INSTALLATION & VM SETUP

1. System Update

sudo pacman -Syu



```
[master][~]$ sudo pacman -Syu
sudo pacman -S qemu-base libvirt virt-manager dnsmasq iptables-nft ebtables openbsd-netcat
[sudo] password for rajat:
:: Synchronizing package databases...
 core                                   123.4 KiB  20.3 KiB/s 00:06 [------------------------------------------------------------] 100%
 extra                                    8.2 MiB   688 KiB/s 00:12 [------------------------------------------------------------] 100%
:: Starting full system upgrade...
resolving dependencies...
looking for conflicting packages...

Package (63)                     Old Version   New Version   Net Change  Download Size

extra/adwaita-cursors            49.0-1        50.0-1          0.00 MiB       0.35 MiB
extra/adwaita-icon-theme         49.0-1        50.0-1          0.02 MiB       0.20 MiB
extra/at-spi2-core               2.58.4-1      2.60.0-1        0.04 MiB       0.57 MiB
extra/atkmm                      2.28.4-1      2.28.4-2        0.00 MiB       0.10 MiB
extra/compiler-rt                22.1.2-1      22.1.3-1        0.00 MiB       4.44 MiB
extra/docker                     1:29.3.1-1    1:29.4.0-1     -0.19 MiB      27.22 MiB
extra/firefox                    149.0-1       149.0.2-1       0.19 MiB      81.33 MiB
extra/gdk-pixbuf2                2.44.4-1      2.44.6-2        0.02 MiB       0.48 MiB
core/glib2                       2.86.5-1      2.88.0-1        0.17 MiB       4.96 MiB
core/glib2-devel                 2.86.5-1      2.88.0-1        0.02 MiB       0.16 MiB
extra/glibmm-2.68                2.86.0-2      2.88.0-1        0.01 MiB       1.07 MiB
extra/glycin                     2.0.8-1       2.1.1-1        -1.41 MiB       3.78 MiB
extra/gnome-keyring              1:48.0-1      1:50.0-1        0.00 MiB       0.70 MiB
extra/gnome-system-monitor       49.1-1        50.0-1          0.18 MiB       0.84 MiB
extra/gsettings-desktop-schemas  49.1-1        50.0-1          0.39 MiB       0.79 MiB
extra/gsettings-system-schemas   49.1-1        50.0-1          0.00 MiB       0.01 MiB
extra/gst-plugins-bad-libs       1.28.1-2      1.28.2-1        0.03 MiB       3.09 MiB
extra/gst-plugins-base           1.28.1-2      1.28.2-1        0.00 MiB       0.21 MiB
extra/gst-plugins-base-libs      1.28.1-2      1.28.2-1        0.04 MiB       2.39 MiB
extra/gst-plugins-good           1.28.1-2      1.28.2-1        0.02 MiB       2.25 MiB
extra/gst-plugins-ugly           1.28.1-2      1.28.2-1        0.00 MiB       0.18 MiB
extra/gstreamer                  1.28.1-2      1.28.2-1        0.05 MiB       2.06 MiB
extra/gtk-update-icon-cache      1:4.20.4-1    1:4.22.2-1      0.00 MiB       0.02 MiB
extra/gtk4                       1:4.20.4-1    1:4.22.2-1      2.02 MiB       9.15 MiB
extra/gtkmm-4.0                  4.20.0-1      4.22.0-1        0.13 MiB       1.50 MiB
extra/gvfs                       1.58.3-1      1.60.0-2        0.00 MiB       0.92 MiB
extra/gvfs-mtp                   1.58.3-1      1.60.0-2        0.00 MiB       0.05 MiB
extra/gweather-locations                       2026.2-1       21.93 MiB       2.78 MiB
core/hwdata                      0.405-1       0.406-1         0.04 MiB       1.68 MiB
```

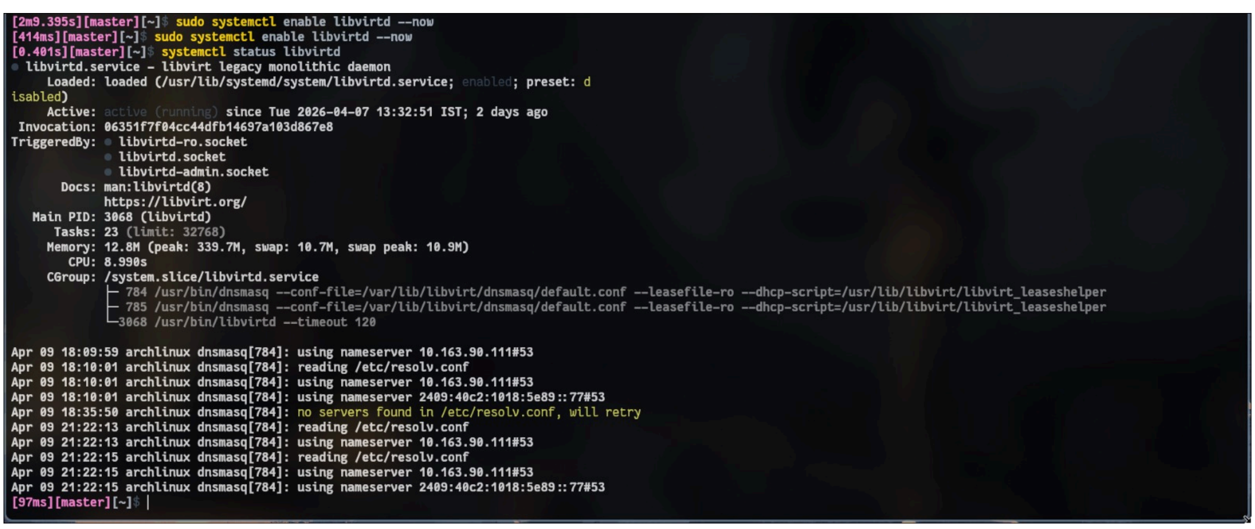
2. Install Required Packages

sudo pacman -S qemu-base libvirt virt-manager dnsmasq iptables-nft ebtables openbsd-netcat

3. Enable libvirt Service

sudo systemctl enable libvirtd --now

systemctl status libvirtd



```
[2m9.395s][master][~]$ sudo systemctl enable libvirtd --now
[414ms][master][~]$ sudo systemctl enable libvirtd --now
[0.401s][master][~]$ systemctl status libvirtd
● libvirtd.service - libvirt legacy monolithic daemon
     Loaded: loaded (/usr/lib/systemd/system/libvirtd.service; enabled; preset: d
isabled)
     Active: active (running) since Tue 2026-04-07 13:32:51 IST; 2 days ago
 Invocation: 06351f7f04cc44dfb14697a103d867e8
TriggeredBy: ● libvirtd-ro.socket
             ● libvirtd.socket
             ● libvirtd-admin.socket
       Docs: man:libvirtd(8)
             https://libvirt.org/
   Main PID: 3068 (libvirtd)
      Tasks: 23 (limit: 32768)
     Memory: 12.8M (peak: 339.7M, swap: 10.7M, swap peak: 10.9M)
        CPU: 8.990s
     CGroup: /system.slice/libvirtd.service
             ├─ 784 /usr/bin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default.conf --leasefile-ro --dhcp-script=/usr/lib/libvirt/libvirt_leaseshelper
             ├─ 785 /usr/bin/dnsmasq --conf-file=/var/lib/libvirt/dnsmasq/default.conf --leasefile-ro --dhcp-script=/usr/lib/libvirt/libvirt_leaseshelper
             └─3068 /usr/bin/libvirtd --timeout 120

Apr 09 18:09:59 archlinux dnsmasq[784]: using nameserver 10.163.90.111#53
Apr 09 18:10:01 archlinux dnsmasq[784]: reading /etc/resolv.conf
Apr 09 18:10:01 archlinux dnsmasq[784]: using nameserver 10.163.90.111#53
Apr 09 18:10:01 archlinux dnsmasq[784]: using nameserver 2409:40c2:1018:5e89::77#53
Apr 09 18:35:50 archlinux dnsmasq[784]: no servers found in /etc/resolv.conf, will retry
Apr 09 21:22:13 archlinux dnsmasq[784]: reading /etc/resolv.conf
Apr 09 21:22:13 archlinux dnsmasq[784]: using nameserver 10.163.90.111#53
Apr 09 21:22:15 archlinux dnsmasq[784]: reading /etc/resolv.conf
Apr 09 21:22:15 archlinux dnsmasq[784]: using nameserver 10.163.90.111#53
Apr 09 21:22:15 archlinux dnsmasq[784]: using nameserver 2409:40c2:1018:5e89::77#53
[97ms][master][~]$ |
```

4. Add User to Required Groups

sudo usermod -aG libvirt $(whoami)

sudo usermod -aG kvm $(whoami)

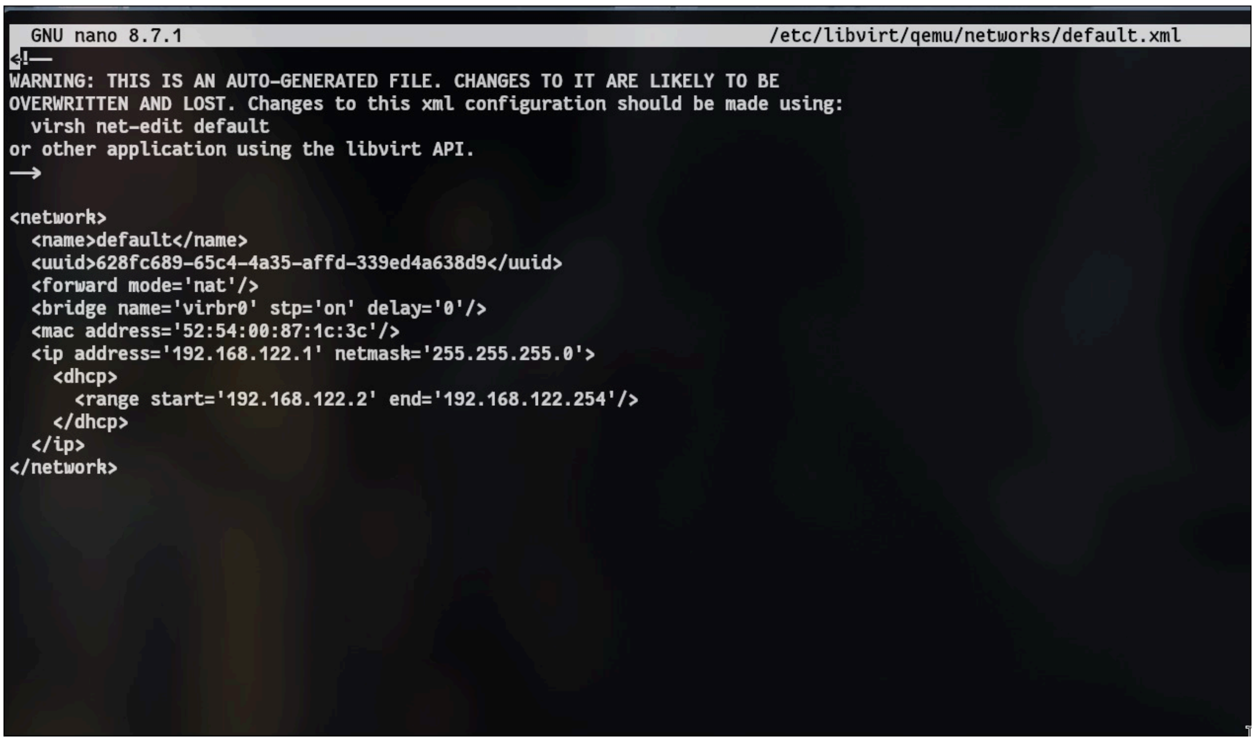
5. Install Kernel & Headers

sudo pacman -S linux linux-headers

sudo reboot

6. Check Bridge Module
lsmod | grep bridge

7.Create Default Network
sudo nano /etc/libvirt/qemu/networks/default.xml




```
<network>
  <name>default</name>
  <forward mode='nat'/>
  <bridge name='virbr0' stp='on' delay='0'/>
  <ip address='192.168.122.1' netmask='255.255.255.0'>
    <dhcp>
      <range start='192.168.122.2' end='192.168.122.254'/>
    </dhcp>
  </ip>
</network>
```

8.Define, Start & Enable Network
sudo virsh net-define /etc/libvirt/qemu/networks/default.xml
sudo virsh net-start default
sudo virsh net-autostart default

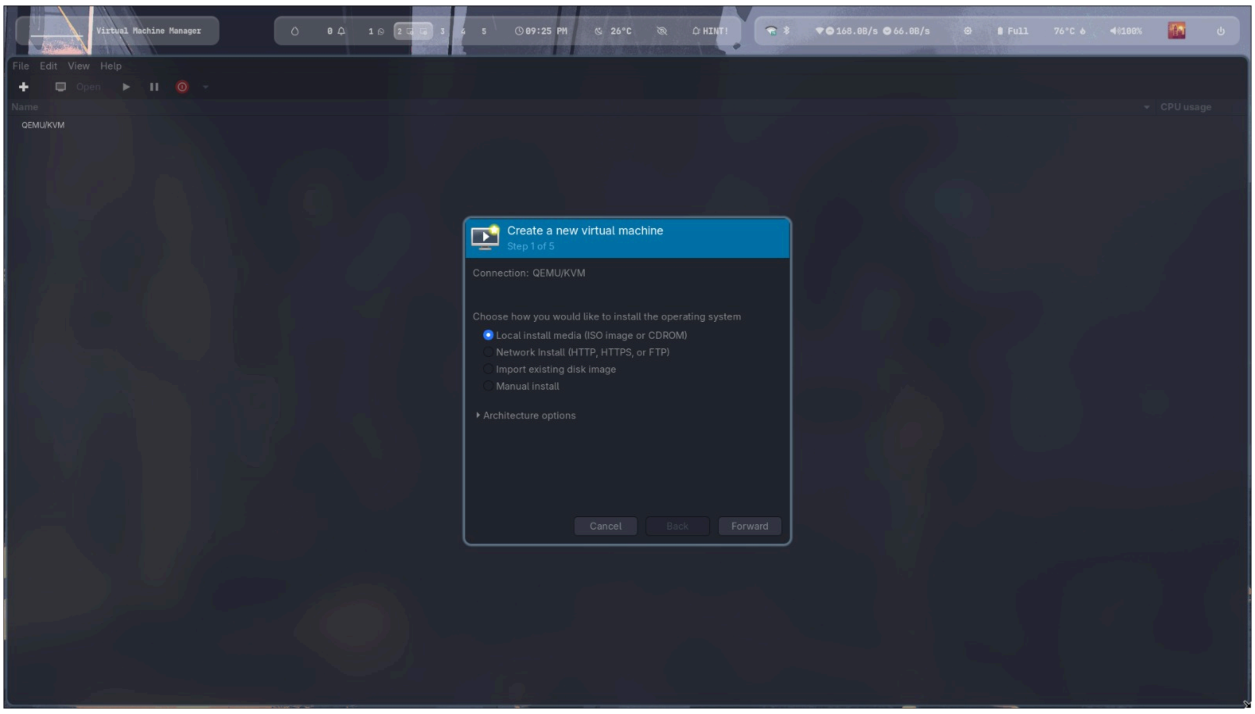
9. Verify Network
ip a | grep virbr0

10.Launch Virtual Machine Manager
virt-manager

VM CREATION STEPS (INSIDE VIRTUAL MACHINE MANAGER)

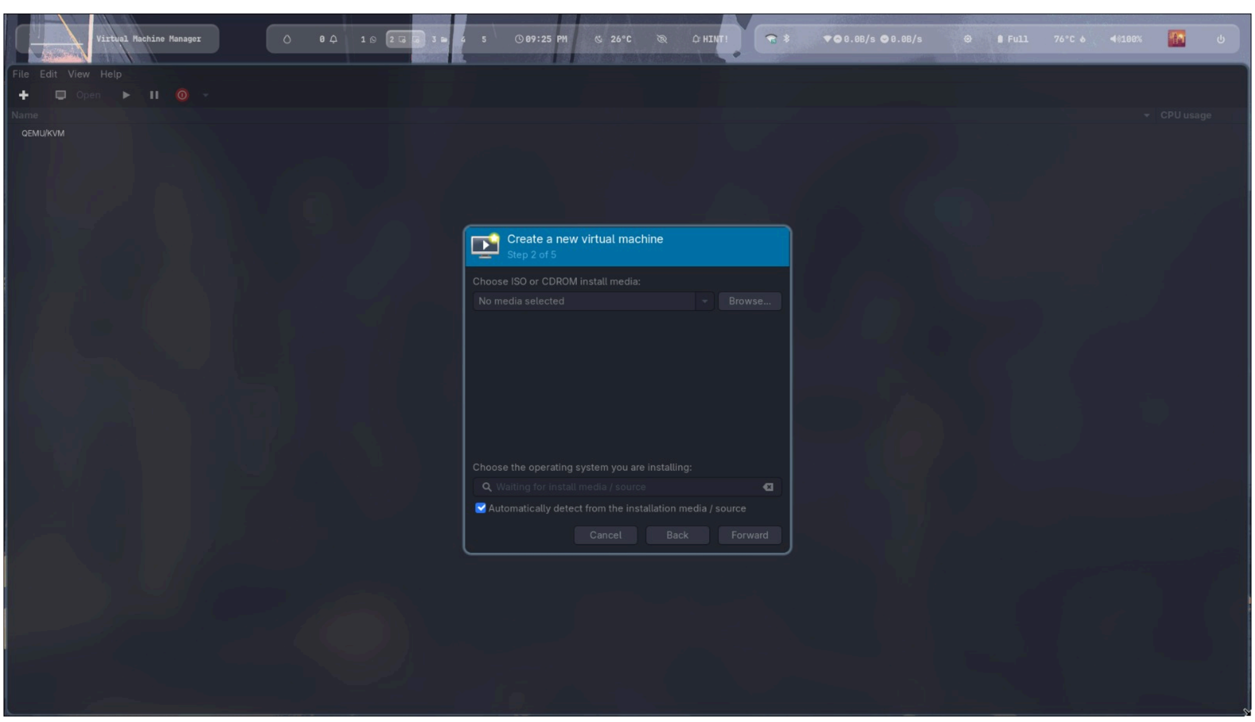
Step 1: Start VM Creation

- **Click "Create a new virtual machine"**



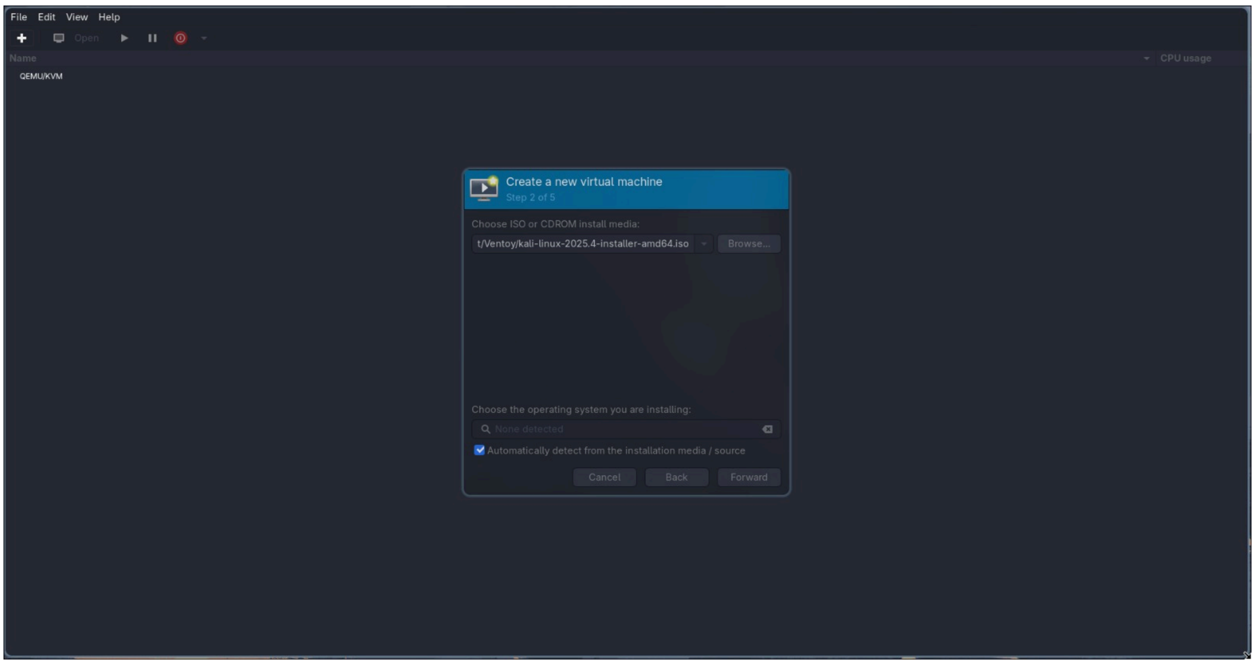
Step 2: Choose Installation Method

- **Select: Local install media (ISO image or CDROM)**
- **Click Forward**



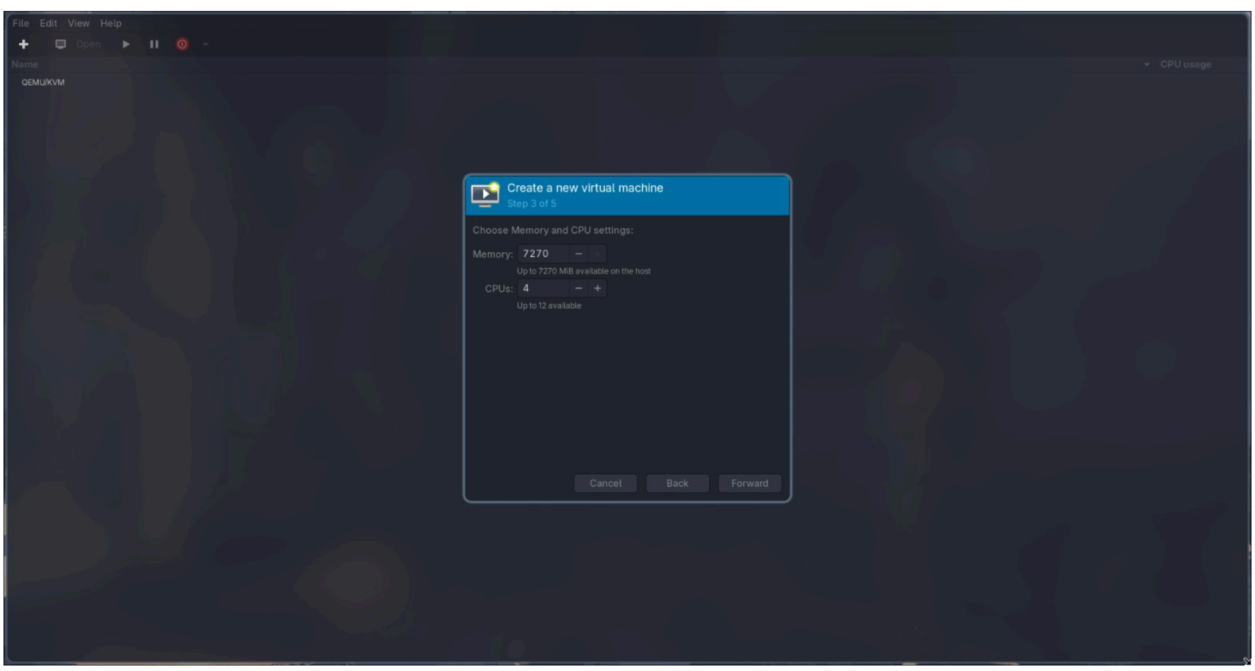
Step 3: Select ISO File

- **Click Browse → Browse Local**
- **Select your OS ISO file (e.g., Ubuntu ISO)**
- **Click Forward**



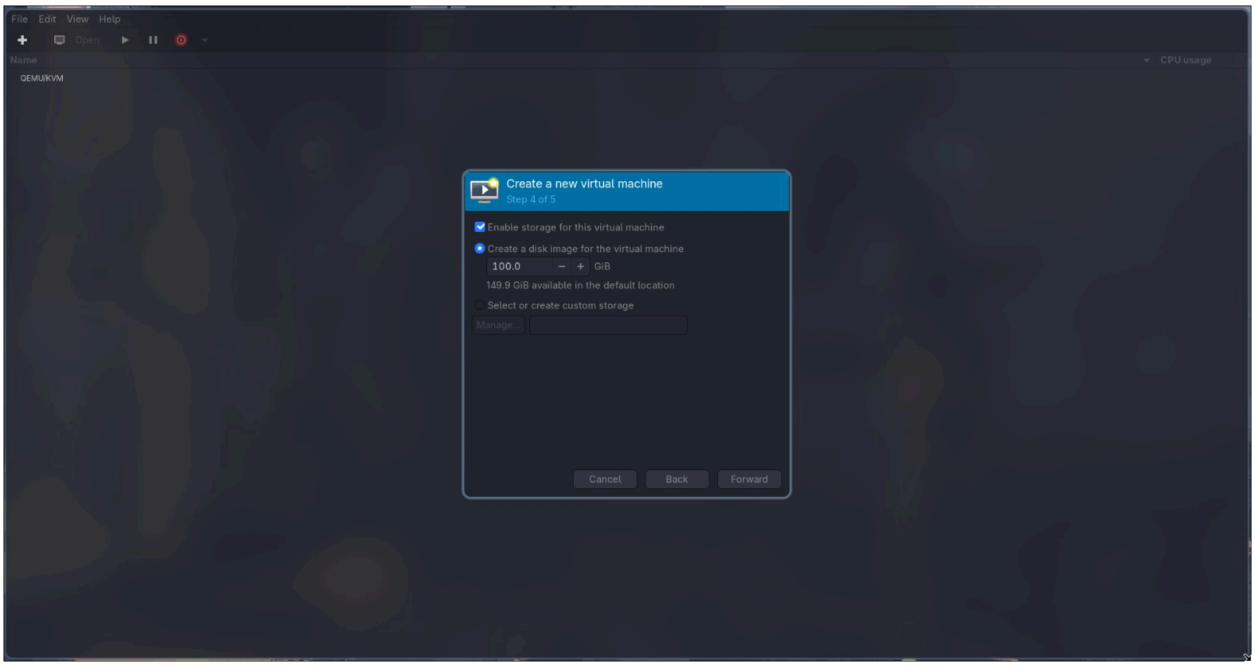
Step 4: Set Memory & CPU

- **RAM: 2048 MB (2 GB)**
- **CPU: 2 cores**
- **Click Forward**



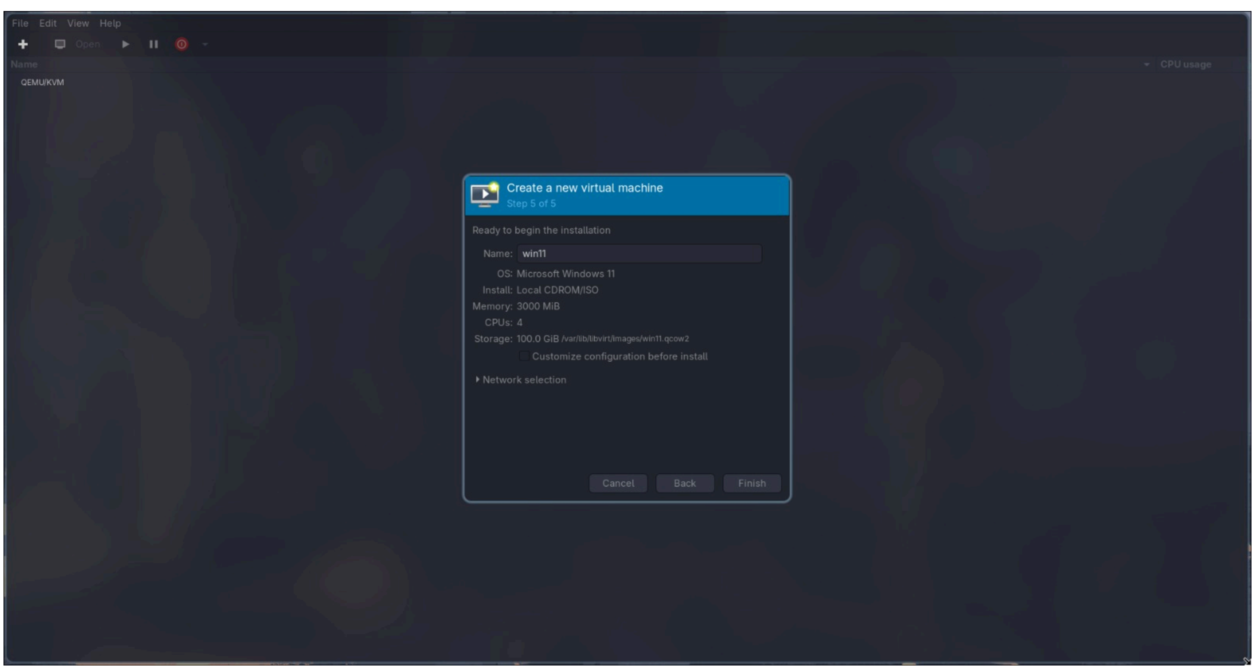
Step 5: Create Disk

- **Select: Create a disk image**
- **Size: 20 GB**
- **Click Forward**



Step 6: Final Configuration

- **Name your VM (e.g., `Ubuntu-VM`)**
- **Ensure network is NAT (default)**
- **Click Finish**



Step 7: Start Installation

- **VM will automatically start**
- **OS installer screen will appear**

Step 8: Install OS

- **Follow on-screen instructions (language, user, password, etc.)**